The sharp A_p bound for the Hardy–Littlewood maximal operator

An exposition of Lerner's proof of Buckley's theorem

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Abstract

B. Muckenhoupt [6] showed that the Hardy–Littlewood maximal operator is bounded on $L^p_w(\mathbb{R}^d)$, $1 < p < \infty$, if and only if $w \in A_p$. S. Buckley [1] proved that its operator norm is at most a constant times $[w]_{A_p}^{1/(p-1)}$, and that this exponent is best possible. We present A. K. Lerner's later proof [5] of this bound in detail, together with Buckley's example showing that the exponent cannot be improved.

1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$, $w(E) := \int_E w(x)\,dx$. Then $E \mapsto w(E)$ is a measure on the Lebesgue measurable sets.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \leq p < \infty$. The weighted L^p space is*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

The central object of this report is the Hardy–Littlewood maximal operator M. Our proofs also use its analogue with respect to the measure $w(x)\,dx$.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

where the supremum is taken over all closed cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. All cubes in this report are of this kind. If the supremum is taken only over the cubes $Q(x,r) := \{y : |y - x|_\infty \leq r/2\}$, $r > 0$, where $|z|_\infty := \max_{1 \leq k \leq d} |z_k|$, the operator is denoted M^c_w. If $w \equiv 1$, the subscript is dropped, giving M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) for $1 < p \leq \infty$ [3, Theorem 1.1]. For $1 < p < \infty$ and a weight w, M is bounded on $L^p_w(\mathbb{R}^d)$ if there is a constant C such that

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w \, dx \right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p \, w \, dx \right)^{1/p} = C \, \|f\|_{L^p_w} \tag{1}$$

for all $f \in L^p_w(\mathbb{R}^d)$.

Remark 1 (The endpoint $p = \infty$)**.** *For $p = \infty$, the boundedness of M has a simple characterisation. Muckenhoupt [6, Theorem 3] shows that, on an interval, M is bounded on L^∞_w if and only if $w > 0$ almost everywhere or $w = 0$ almost everywhere. This condition only concerns where w vanishes.*

Remark 2 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\int_{\mathbb{R}^d} |f|^p \, w \, dx = 0$ for every measurable function f. Applying this to f and to Mf gives $\|Mf\|_{L^p_w} = 0 = \|f\|_{L^p_w}$, so (1) holds with any constant C.*

If (1) holds, then each $f \in L^p_w(\mathbb{R}^d)$ and each cube Q give a condition on w, and Muckenhoupt showed that a suitable choice of f for each cube gives the following necessary condition. It is not needed for the proof of the main result, but it motivates the definition of A_p below, so its proof is included in Appendix A.

Proposition 1 ([6, Theorem 9], [3, proof of Theorem 1.7])**.** *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then every cube Q satisfies $\int_Q w^{-\frac{1}{p-1}} \, dx < \infty$ and*

$$\left(\frac{1}{|Q|} \int_Q w \, dx \right) \left(\frac{1}{|Q|} \int_Q w^{-\frac{1}{p-1}} \, dx \right)^{p-1} \leq C^p. \tag{2}$$

In particular, $w > 0$ almost everywhere.

Definition 4 (Muckenhoupt A_p weights [6, (7.10)])**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|} \int_Q w(x) \, dx \right) \left(\frac{1}{|Q|} \int_Q w(x)^{-\frac{1}{p-1}} \, dx \right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. The weight w belongs to A_p if $[w]_{A_p} < \infty$.

Definition 5 (Dual weight [1, Section 1])**.** *Let w be a weight and let $1 < p < \infty$. The* dual weight *of w is $\sigma := w^{-\frac{1}{p-1}}$, with $\sigma = \infty$ where $w = 0$. For measurable sets E, $\sigma(E) := \int_E \sigma \, dx$. With this notation (3) takes the more compact form*

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|} \left(\frac{\sigma(Q)}{|Q|} \right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then $w \in A_p$ with*

$$[w]_{A_p} \leq C^p. \tag{4}$$

Muckenhoupt proved that the condition $w \in A_p$ is also sufficient.

Theorem 1 (Muckenhoupt [6, Theorem 9]). *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem 1 shows that the operator norm of M on $L^p_w(\mathbb{R}^d)$, that is, the smallest constant C in (1), is finite precisely when $w \in A_p$. By Corollary 1, this norm is bounded from below by $[w]_{A_p}^{1/p}$. The natural question is therefore whether it is also bounded from above by a power of $[w]_{A_p}$, and which power is the smallest possible. This was answered by Buckley.

Theorem 2 (Buckley [1, Theorem 2.5]). *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \le c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}.$$

Proposition 2 (Buckley [1, Theorem 2.5]). *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant c such that $\|Mf\|_{L^p_w} \le c\, [w]_{A_p}^{\alpha}\, \|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

Buckley's proof of Theorem 2 [1, proof of Theorem 2.5], like Muckenhoupt's proof of Theorem 1 [6, Lemma 5 and proof of Theorem 9], relies on a deep property that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$ [3, Theorem 4.5], together with interpolation. Lerner [5, Theorem C] found a short and elementary proof of Theorem 2. This report presents his argument in detail and completes it with a proof of Proposition 2, based on Buckley's example in $\mathbb{R}$ [1, proof of Theorem 2.5] and extended here to $\mathbb{R}^d$.

Remark 3 (An alternative proof of Theorem 1). *If M is bounded on $L^p_w(\mathbb{R}^d)$, then $w \in A_p$ by Corollary 1, which follows from Proposition 1. Conversely, if $w \in A_p$, then M is bounded on $L^p_w(\mathbb{R}^d)$ by Theorem 2. Combined, these results imply Theorem 1.*

2 Proofs

2.1 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z - y|_\infty \le \ell(Q)$, since both z and y lie in Q. Hence $Q \subseteq Q(y, 2\ell(Q))$. If c is the centre of Q, then $|y - c|_\infty \le \frac{1}{2}\ell(Q)$, so every $z \in Q(y, 2\ell(Q))$ satisfies $|z - c|_\infty \le |z - y|_\infty + |y - c|_\infty \le \frac{3}{2}\ell(Q)$, that is, $z \in 3Q$. □

Lemma 2 ([4, Section 2.1.1]). *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \le 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and

$|Q(x, 2\ell(Q))| = 2^d|Q|$ gives

$$\begin{aligned} Mf(x) &= \sup_{Q\ni x} \frac{1}{|Q|} \int_Q |f(y)|\,dy \\ &\le \sup_{Q\ni x} \frac{2^d}{|Q(x,2\ell(Q))|} \int_{Q(x,2\ell(Q))} |f(y)|\,dy \\ &\le 2^d \sup_{r>0} \frac{1}{|Q(x,r)|} \int_{Q(x,r)} |f(y)|\,dy \\ &= 2^d M^c f(x). \end{aligned}$$

□

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere and σ, as in Definition 5, is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q, and $L^p_w(\mathbb{R}^d) \subseteq L^1_{\mathrm{loc}}(\mathbb{R}^d)$.*

Proof. Let Q be a cube with $w(Q) > 0$. By the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \le [w]_{A_p} \frac{|Q|}{w(Q)} < \infty, \tag{5}$$

so $\sigma(Q) < \infty$ on any cube Q where $w(Q) > 0$.

Assume, for contradiction, that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, $w > 0$ on a set F with $|F| > 0$, so some cube Q satisfies $|Q \cap E| > 0$ and $|Q \cap F| > 0$. Then $w(Q) \ge w(Q \cap F) > 0$, so $\sigma(Q) < \infty$ by (5). But $\sigma = \infty$ on $Q \cap E$ by Definition 5, so $\sigma(Q) = \infty$, a contradiction. Therefore $w > 0$ almost everywhere. In particular $w(Q) > 0$, and hence $\sigma(Q) < \infty$ by (5), for every cube Q.

Every cube Q has $w(Q) < \infty$, so $w < \infty$ almost everywhere. Hence $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. As every compact set lies in a cube, σ is locally integrable and hence a weight. Finally, for $f \in L^p_w(\mathbb{R}^d)$ and every cube Q, applying Hölder's inequality gives

$$\int_Q |f|\,dx = \int_Q |f|\, w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\,dx \le \Big(\int_Q |f|^p\, w\,dx\Big)^{\frac{1}{p}} \Big(\int_Q w^{-\frac{1}{p-1}}\,dx\Big)^{\frac{p-1}{p}} \le \|f\|_{L^p_w}\, \sigma(Q)^{\frac{p-1}{p}},$$

which is finite. □

Remark 4 (The Vitali covering lemma)**.** *In the proof of the weak type $(1,1)$ estimate for M [3, Theorem 1.1], a compact set $K \subseteq \{Mf > t\}$ is covered by finitely many cubes Q with $\frac{1}{|Q|}\int_Q |f|\,dx > t$. The Vitali covering lemma [3, Lemma A.4], which holds for balls as well as cubes, selects disjoint cubes $Q_1, \dots, Q_N$ among them with $K \subseteq \bigcup_j 3Q_j$. Since $|3Q_j| = 3^d|Q_j|$, this gives*

$$|K| \le \sum_{j=1}^N |3Q_j| = 3^d \sum_{j=1}^N |Q_j| \le \frac{3^d}{t} \int_{\mathbb{R}^d} |f|\,dx.$$

For the measure $v(x)\,dx$, the equality $|3Q_j| = 3^d|Q_j|$ would have to be replaced by an inequality $v(3Q_j) \le C\, v(Q_j)$, and no constant C works for all weights v [3, Remark 1.11]. For example, if $v = \varepsilon$ on Q_j and $v = 1$ on the rest of $3Q_j$, then $v(3Q_j)/v(Q_j) \ge (3^d - 1)/\varepsilon$, which is arbitrarily large. The Besicovitch covering lemma covers K by the selected cubes themselves, without enlarging them, and therefore gives a constant independent of v.

Lemma 4 (Besicovitch covering lemma [2, Theorem 1.1])**.** *Let $E \subset \mathbb{R}^d$ be a bounded set, and suppose that for each $x \in E$ a cube $Q(x, r_x)$ centred at x is given. Then there are points $x_1, x_2, \ldots \in E$ such that the cubes $Q(x_j, r_{x_j})$ cover E and every point of $\mathbb{R}^d$ lies in at most N_d of them, where N_d depends only on d.*

Proof. Assume that E is nonempty and let $D := \sup_{x,y\in E} |x - y|_\infty < \infty$. If $r_x \geq 2D$ for some $x \in E$, then $E \subseteq Q(x, r_x)$ and this single cube suffices. Hence assume that $r_x < 2D$ for every $x \in E$.

Let $s_0 := \sup_{x\in E} r_x$ and choose $x_1 \in E$ with $r_{x_1} > s_0/2$. Given $x_1, \ldots, x_m$, let

$$s_m := \sup \Big\{ r_x : x \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j}) \Big\}$$

and choose $x_{m+1} \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})$ with $r_{x_{m+1}} > s_m/2$. The selection stops if this set is empty. If $j < i$, then x_i lies outside $Q(x_j, r_{x_j})$, and x_i was a possible choice when x_j was chosen. Hence

$$|x_i - x_j|_\infty > \frac{r_{x_j}}{2} \qquad \text{and} \qquad r_{x_i} \leq s_{j-1} < 2r_{x_j}. \tag{6}$$

If the selection stops, the cubes $Q(x_j, r_{x_j})$ cover E by construction. Assume that it does not stop, and suppose, for contradiction, that some $x \in E$ is not covered. Then x is a possible choice at every step, so $r_x \leq s_{j-1} < 2r_{x_j}$ for every j. If $j < i$, then by (6)

$$\frac{r_{x_i}}{8} + \frac{r_{x_j}}{8} < \frac{3r_{x_j}}{8} < |x_i - x_j|_\infty,$$

so the cubes $Q(x_j, r_{x_j}/4)$ are pairwise disjoint. Since $x_j \in E$ and $r_{x_j} < 2D$, they are all contained in $Q(x_1, 4D)$. Comparing volumes, for every N,

$$N \left(\frac{r_x}{8}\right)^d < \sum_{j=1}^{N} \left(\frac{r_{x_j}}{4}\right)^d \leq (4D)^d,$$

which is false for large N. Hence every $x \in E$ is covered.

Fix $z \in \mathbb{R}^d$. For $\varepsilon \in \{-1, 1\}^d$ let

$$O_\varepsilon := \{y \in \mathbb{R}^d : \varepsilon_k(y_k - z_k) \geq 0 \text{ for } k = 1, \ldots, d\}.$$

Every $y \in \mathbb{R}^d$ lies in O_ε with $\varepsilon_k := 1$ if $y_k \geq z_k$ and $\varepsilon_k := -1$ otherwise, so these 2^d orthants cover $\mathbb{R}^d$. If $y, y' \in O_\varepsilon$, then $y_k - z_k$ and $y'_k - z_k$ have the same sign for every k, and hence

$$|y - y'|_\infty \leq \max\{|y - z|_\infty,\ |y' - z|_\infty\}. \tag{7}$$

For each ε let

$$J_\varepsilon := \{x_i : x_i \in O_\varepsilon \text{ and } z \in Q(x_i, r_{x_i})\}.$$

Fix ε_0 with J_{ε_0} nonempty and let $j := \min\{i : x_i \in J_{\varepsilon_0}\}$, so that x_j is the point of J_{ε_0} that was chosen first. Let $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$, so that $i > j$. Since z lies in both $Q(x_i, r_{x_i})$ and $Q(x_j, r_{x_j})$, (6) and (7) give

$$\frac{r_{x_j}}{2} < |x_i - x_j|_\infty \leq \max\{|x_i - z|_\infty,\ |x_j - z|_\infty\} \leq \max\Big\{\frac{r_{x_i}}{2},\ \frac{r_{x_j}}{2}\Big\}.$$

Hence $r_{x_i} > r_{x_j}$, and by (6) also $|x_i - z|_\infty \le r_{x_i}/2 < r_{x_j}$.

For each $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$ consider the cube $Q(x_i, r_{x_j}/2)$. If $x_i, x_m \in J_{\varepsilon_0} \setminus \{x_j\}$ with $m < i$, then $r_{x_m} > r_{x_j}$, so (6) gives $|x_i - x_m|_\infty > r_{x_m}/2 > r_{x_j}/2$. Thus the centres of these cubes are more than their side length apart, and the cubes are pairwise disjoint. Moreover, for $x_i \in J_{\varepsilon_0} \setminus \{x_j\}$ every $y \in Q(x_i, r_{x_j}/2)$ satisfies

$$|y - z|_\infty \le |y - x_i|_\infty + |x_i - z|_\infty < \frac{r_{x_j}}{4} + r_{x_j} = \frac{5r_{x_j}}{4},$$

so all of them are contained in $Q(z, 5r_{x_j}/2)$. If n of these cubes are chosen, then, since they are disjoint, their total volume is at most the volume of $Q(z, 5r_{x_j}/2)$, that is,

$$n\left(\frac{r_{x_j}}{2}\right)^d \le \left(\frac{5r_{x_j}}{2}\right)^d = 5^d\left(\frac{r_{x_j}}{2}\right)^d.$$

Hence $n \le 5^d$, so J_{ε_0} has at most $1 + 5^d$ points. Since ε_0 was an arbitrary choice with J_{ε_0} nonempty, every J_ε has at most $1 + 5^d$ points. Every cube $Q(x_i, r_{x_i})$ containing z has its centre in some O_ε, and hence in J_ε. Summing over the 2^d orthants, z lies in at most $2^d(1 + 5^d)$ of the cubes $Q(x_i, r_{x_i})$, and the lemma holds with $N_d := 2^d(1 + 5^d)$. □

Lemma 5 ([3, Theorem 1.10])**.** *Let v be a weight with $v(Q) > 0$ for every cube Q, and let f be a measurable function with $|f|\, v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $M^c_v f$ is measurable, and for every $t > 0$,*

$$v\big(\{x \in \mathbb{R}^d : M^c_v f(x) > t\}\big) \le \frac{N_d}{t}\,\|f\|_{L^1_v},$$

where N_d is the constant in Lemma 4.

Proof. Let $r > 0$ and $x \in \mathbb{R}^d$, and let (x_n) be a sequence converging to x. If $|y - x|_\infty < r/2$, then $|y - x_n|_\infty < r/2$ for all large n. If $|y - x|_\infty > r/2$, then $|y - x_n|_\infty > r/2$ for all large n. Hence $\chi_{Q(x_n,r)}(y) \to \chi_{Q(x,r)}(y)$ for every y outside the boundary of $Q(x, r)$, which is a set of measure zero. Moreover, $R := \sup_n |x_n - x|_\infty$ is finite, and $Q(x_n, r) \subseteq Q(x, r + 2R)$ for every n, so $\chi_{Q(x_n,r)}\,|f|\, v \le \chi_{Q(x,r+2R)}\,|f|\, v$, which is integrable since $|f|\, v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. By dominated convergence,

$$\lim_{n\to\infty}\int_{Q(x_n,r)} |f|\, v\, dy = \lim_{n\to\infty}\int_{\mathbb{R}^d} \chi_{Q(x_n,r)}\,|f|\, v\, dy = \int_{\mathbb{R}^d} \chi_{Q(x,r)}\,|f|\, v\, dy = \int_{Q(x,r)} |f|\, v\, dy.$$

Thus $x \mapsto \int_{Q(x,r)} |f|\, v\, dy$ is sequentially continuous, and hence continuous. Taking $f \equiv 1$ shows that $x \mapsto v(Q(x, r))$ is continuous as well. Since $v(Q(x, r)) > 0$, the average

$$x \mapsto \frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f|\, v\, dy$$

is continuous for every $r > 0$. For $t > 0$ the set $\{M^c_v f > t\}$ is the union over $r > 0$ of the sets where this average exceeds t. These are open, so $\{M^c_v f > t\}$ is open. Since $t > 0$ was arbitrary and $M^c_v f \ge 0$, the function $M^c_v f$ is measurable.

Now fix $t > 0$. For $k = 1, 2, \ldots$ let $E_k := \{x : |x| < k \text{ and } M^c_v f(x) > t\}$. Let $x \in E_k$. Since $M^c_v f(x) > t$, the definition of M^c_v gives some $r > 0$ with $\frac{1}{v(Q(x,r))}\int_{Q(x,r)} |f|\, v\, dy > t$.

Choose such an r and call it r_x. Since E_k is bounded, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover E_k and $\sum_j \chi_{Q(x_j,r_{x_j})} \le N_d$. Hence

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{t} \sum_j \int_{Q(x_j,r_{x_j})} |f|\, v\, dy \\ &= \frac{1}{t} \int_{\mathbb{R}^d} \Big(\sum_j \chi_{Q(x_j,r_{x_j})} \Big) |f|\, v\, dy \le \frac{N_d}{t}\, \|f\|_{L^1_v}, \end{aligned}$$

where the second inequality holds by the choice of r_{x_j}, and the equality follows from monotone convergence. Since $E_1 \subseteq E_2 \subseteq \cdots$ and v is a measure, the bound for $v(E_k)$ gives

$$v\big(\{M^c_v f > t\}\big) = v\Big(\bigcup_k E_k \Big) = \lim_{k\to\infty} v(E_k) \le \frac{N_d}{t}\, \|f\|_{L^1_v}.$$

□

Lemma 6 ([3, Theorem 1.10])**.** *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M^c_v f\|_{L^q_v} \le C_{q,d}\, \|f\|_{L^q_v}$$

for all $f \in L^q_v(\mathbb{R}^d)$.

Proof. Let $f \in L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, so that $f = f_1 + f_2$ with $f_1 \in L^1_v(\mathbb{R}^d)$ and $f_2 \in L^\infty_v(\mathbb{R}^d)$. By the definition of $\|f_2\|_{L^\infty_v}$, the sets $\{|f_2| > \|f_2\|_{L^\infty_v} + 1/n\}$, $n = 1, 2, \ldots$, have v-measure zero, and hence so does their union $\{|f_2| > \|f_2\|_{L^\infty_v}\}$. This implies

$$\int_Q |f_2|\, v\, dy \le \|f_2\|_{L^\infty_v}\, v(Q) \quad \text{for every cube } Q, \tag{8}$$

and therefore

$$\int_Q |f|\, v\, dy \le \int_Q |f_1|\, v\, dy + \int_Q |f_2|\, v\, dy \le \|f_1\|_{L^1_v} + \|f_2\|_{L^\infty_v}\, v(Q) < \infty.$$

Hence $|f|\, v \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, so $M^c_v f$ is defined and, by Lemma 5, measurable.

Let $f, h \in L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, $x \in \mathbb{R}^d$ and $r > 0$. The pointwise inequality $|f + h| \le |f| + |h|$ gives

$$\begin{aligned} \frac{1}{v(Q(x,r))} \int_{Q(x,r)} |f + h|\, v\, dy &\le \frac{1}{v(Q(x,r))} \int_{Q(x,r)} |f|\, v\, dy + \frac{1}{v(Q(x,r))} \int_{Q(x,r)} |h|\, v\, dy \\ &\le M^c_v f(x) + M^c_v h(x). \end{aligned}$$

The right-hand side does not depend on r, so taking the supremum over $r > 0$ yields $M^c_v(f + h)(x) \le M^c_v f(x) + M^c_v h(x)$. For a scalar λ,

$$\begin{aligned} M^c_v(\lambda f)(x) &= \sup_{r>0} \frac{1}{v(Q(x,r))} \int_{Q(x,r)} |\lambda|\, |f|\, v\, dy \\ &= |\lambda| \sup_{r>0} \frac{1}{v(Q(x,r))} \int_{Q(x,r)} |f|\, v\, dy = |\lambda|\, M^c_v f(x), \end{aligned}$$

where the second equality holds since $|\lambda| \geq 0$. Hence M_v^c is sublinear.

By Lemma 5, $v(\{M_v^c f > t\}) \leq \frac{N_d}{t}\|f\|_{L_v^1}$ for every $f \in L_v^1(\mathbb{R}^d)$ and $t > 0$. For $f \in L_v^\infty(\mathbb{R}^d)$, (8) with $f_2 = f$ shows that every average in the definition of $M_v^c f(x)$ is at most $\|f\|_{L_v^\infty}$. Hence $M_v^c f(x) \leq \|f\|_{L_v^\infty}$ for every $x \in \mathbb{R}^d$, and in particular $\|M_v^c f\|_{L_v^\infty} \leq \|f\|_{L_v^\infty}$.

The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(x)\,dx)$ with exponents 1 and ∞, now gives the claim with a constant depending only on q and on the constants N_d and 1 in these two estimates, and hence only on q and d. □

2.2 Proof of Theorem 2

Lemma 7 (Lerner [5, proof of Theorem C])**.** *Let $1 < p < \infty$, let $w \in A_p$ be a weight that is not almost everywhere zero, and let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then, for every $x \in \mathbb{R}^d$,*

$$M^c f(x) \leq 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}} \Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{9}$$

Proof. By Lemma 3, $w > 0$ almost everywhere and σ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q, so M_w^c and M_σ^c are defined. Since $|f|\,\sigma^{-1}\sigma = |f| \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, Lemma 5 with $v = \sigma$ shows that $M_\sigma^c(|f|\sigma^{-1})$ is measurable.

For $x \in \mathbb{R}^d$, $r > 0$ and $y \in Q(x,r)$, Lemma 1 gives $Q(x,r) \subseteq Q(y,2r) \subseteq Q(x,3r)$, so

$$\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)} |f|\,dz \leq \frac{1}{\sigma(Q(y,2r))}\int_{Q(y,2r)} |f|\,\sigma^{-1}\,\sigma\,dz \leq M_\sigma^c\big(|f|\sigma^{-1}\big)(y). \tag{10}$$

The inequality $w(Q(x,r)) \leq w(Q(x,3r))$, the equality $|Q(x,3r)| = 3^d\,|Q(x,r)|$ and the definition of $[w]_{A_p}$ imply

$$\frac{w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}}{|Q(x,r)|^p} \leq 3^{dp}\,\frac{w(Q(x,3r))\,\sigma(Q(x,3r))^{p-1}}{|Q(x,3r)|^p} \leq 3^{dp}\,[w]_{A_p}. \tag{11}$$

Multiplying and dividing by $w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}$, and then bounding the two marked factors by (11) and (10), gives

$$\begin{aligned}
&\left(\frac{1}{|Q(x,r)|}\int_{Q(x,r)} |f|\,dz\right)^{p-1} \\
&\quad= \underbrace{\frac{w(Q(x,r))\,\sigma(Q(x,3r))^{p-1}}{|Q(x,r)|^p}}_{\leq 3^{dp}[w]_{A_p} \text{ by (11)}} \cdot \frac{1}{w(Q(x,r))}\int_{Q(x,r)} \left(\underbrace{\frac{1}{\sigma(Q(x,3r))}\int_{Q(x,r)} |f|\,dz}_{\leq M_\sigma^c(|f|\sigma^{-1})(y) \text{ by (10)}}\right)^{p-1} dy \\
&\quad\leq 3^{dp}\,[w]_{A_p} \cdot \frac{1}{w(Q(x,r))}\int_{Q(x,r)} \Big(M_\sigma^c\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy \\
&\quad\leq 3^{dp}\,[w]_{A_p}\,M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x).
\end{aligned}$$

Raising both sides to the power $\frac{1}{p-1}$ and taking the supremum over $r > 0$ gives (9). □

Proof of Theorem 2. Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L_w^p(\mathbb{R}^d)$. For $w = 0$ almost everywhere, the integrands in $\|Mf\|_{L_w^p}$ and $\|f\|_{L_w^p}$ vanish almost everywhere, so $\|Mf\|_{L_w^p} = 0 = \|f\|_{L_w^p}$ and the inequality holds trivially. Assume therefore that w is not

almost everywhere zero. Lemma 3 gives $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and $w(Q) > 0$ and $\sigma(Q) > 0$ for every cube Q. Lemma 6 therefore shows that M^c_w and M^c_σ are of strong type (q,q) on $L^q_w(\mathbb{R}^d)$ and $L^q_\sigma(\mathbb{R}^d)$, respectively, for every $1 < q < \infty$.

The identities $w^{1-\frac{p}{p-1}} = \sigma$ and $\sigma^{1-p} = w$, together with Lemma 6 with $v = \sigma$ and $q = p$, give

$$\left\| \left(M^c_\sigma(|f|\sigma^{-1})\right)^{p-1} w^{-1} \right\|^{\frac{1}{p-1}}_{L^{p/(p-1)}_w} = \left\| M^c_\sigma\left(|f|\sigma^{-1}\right) \right\|_{L^p_\sigma} \le C_{p,d} \left\| |f|\sigma^{-1} \right\|_{L^p_\sigma} = C_{p,d} \|f\|_{L^p_w}. \quad (12)$$

In particular $\left(M^c_\sigma(|f|\sigma^{-1})\right)^{p-1} w^{-1} \in L^{p/(p-1)}_w(\mathbb{R}^d)$. Lemma 2 gives the first inequality below, and bounding the marked factors as indicated gives

$$\begin{aligned}
\|Mf\|_{L^p_w} &\le 2^d \Big\| \underbrace{M^c f}_{\text{bounded by Lemma 7}} \Big\|_{L^p_w} \\
&\le 2^d \left\| 3^{\frac{dp}{p-1}} [w]^{\frac{1}{p-1}}_{A_p} \left(M^c_w \Big(\left(M^c_\sigma(|f|\sigma^{-1})\right)^{p-1} w^{-1} \Big) \right)^{\frac{1}{p-1}} \right\|_{L^p_w} \\
&= 2^d\, 3^{\frac{dp}{p-1}} [w]^{\frac{1}{p-1}}_{A_p} \underbrace{\left\| M^c_w \Big(\left(M^c_\sigma(|f|\sigma^{-1})\right)^{p-1} w^{-1} \Big) \right\|^{\frac{1}{p-1}}_{L^{p/(p-1)}_w}}_{\text{bounded by Lemma 6 with } v=w,\ q=\frac{p}{p-1}} \\
&\le 2^d\, 3^{\frac{dp}{p-1}} C^{\frac{1}{p-1}}_{\frac{p}{p-1},d} [w]^{\frac{1}{p-1}}_{A_p} \underbrace{\left\| \left(M^c_\sigma(|f|\sigma^{-1})\right)^{p-1} w^{-1} \right\|^{\frac{1}{p-1}}_{L^{p/(p-1)}_w}}_{\text{bounded by (12)}} \\
&\le 2^d\, 3^{\frac{dp}{p-1}} C^{\frac{1}{p-1}}_{\frac{p}{p-1},d} [w]^{\frac{1}{p-1}}_{A_p} C_{p,d} \|f\|_{L^p_w}.
\end{aligned}$$

□

2.3 Sharpness of the exponent

Proof of Proposition 2. Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant c such that

$$\|Mf\|_{L^p_w} \le c\, [w]^\alpha_{A_p} \|f\|_{L^p_w} \quad (13)$$

for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$. Let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere by Lemma 3, and for every cube Q, Hölder's inequality with exponents p and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|} \int_Q w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\, dy \le \left(\frac{1}{|Q|} \int_Q w\, dy \right)^{\frac{1}{p}} \left(\frac{1}{|Q|} \int_Q w^{-\frac{1}{p-1}}\, dy \right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Together with Remark 2, this shows that (13) remains true if α is replaced by $\max\{\alpha, 0\}$, so α may be assumed to satisfy $0 \le \alpha < \frac{1}{p-1}$. For $0 < \delta < 1$ let

$$w_\delta(x) := |x|^{(d-\delta)(p-1)} \qquad \text{and} \qquad f_\delta(x) := w_\delta(x)^{-\frac{1}{p-1}} \chi_{Q(0,2)}(x) = |x|^{\delta-d} \chi_{Q(0,2)}(x).$$

For every $r > 0$, polar coordinates give

$$\int_{B(0,r)} w_\delta(y)^{-\frac{1}{p-1}}\, dy = \int_{B(0,r)} |y|^{\delta-d}\, dy = |S^{d-1}| \int_0^r s^{\delta-d}\, s^{d-1}\, ds = |S^{d-1}|\, \frac{r^\delta}{\delta}, \quad (14)$$

where $|S^{d-1}|$ denotes the surface measure of the unit sphere $S^{d-1} \subseteq \mathbb{R}^d$.

The function w_δ is continuous and positive outside the origin, so it is a weight that is not almost everywhere zero. Let $Q(c,\ell)$ be a cube. Every $y \in Q(c,\ell)$ satisfies $|y-c| \le \sqrt{d}\,|y-c|_\infty \le \frac{\sqrt{d}}{2}\ell$.

If $|c| \le \sqrt{d}\,\ell$, then $Q(c,\ell) \subseteq B(0, 2\sqrt{d}\,\ell)$. On this ball $w_\delta \le (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}$, and (14) gives

$$\begin{aligned}
&\left(\frac{1}{|Q(c,\ell)|}\int_{Q(c,\ell)} w_\delta\,dy\right)\left(\frac{1}{|Q(c,\ell)|}\int_{Q(c,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \\
&\quad\le (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}\left(\frac{1}{\ell^d}\int_{B(0,2\sqrt{d}\,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \\
&\quad= (2\sqrt{d}\,\ell)^{(d-\delta)(p-1)}\left(\frac{|S^{d-1}|\,(2\sqrt{d}\,\ell)^{\delta}}{\delta\,\ell^d}\right)^{p-1} = (2\sqrt{d})^{d(p-1)}\,|S^{d-1}|^{p-1}\,\delta^{1-p}.
\end{aligned}$$

If $|c| > \sqrt{d}\,\ell$, then $|y-c| < \frac{1}{2}|c|$, so $\frac{1}{2}|c| \le |y| \le \frac{3}{2}|c|$ for every $y \in Q(c,\ell)$. Bounding w_δ by $\left(\frac{3}{2}|c|\right)^{(d-\delta)(p-1)}$ and $|y|^{\delta-d}$ by $\left(\frac{1}{2}|c|\right)^{\delta-d}$ on $Q(c,\ell)$ gives

$$\left(\frac{1}{|Q(c,\ell)|}\int_{Q(c,\ell)} w_\delta\,dy\right)\left(\frac{1}{|Q(c,\ell)|}\int_{Q(c,\ell)} |y|^{\delta-d}\,dy\right)^{p-1} \le 3^{(d-\delta)(p-1)} \le 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes gives $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le C\,\delta^{1-p}, \qquad C := \max\{(2\sqrt{d})^{d(p-1)}|S^{d-1}|^{p-1},\, 3^{d(p-1)}\}. \tag{15}$$

The norm of f_δ satisfies

$$\|f_\delta\|_{L^p_{w_\delta}}^p = \int_{Q(0,2)} \underbrace{|y|^{(\delta-d)p}}_{|f_\delta(y)|^p}\,\underbrace{|y|^{(d-\delta)(p-1)}}_{w_\delta(y)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \le \int_{B(0,2\sqrt{d})} |y|^{\delta-d}\,dy < \infty,$$

by $Q(0,2) \subseteq B(0,2\sqrt{d})$ and (14). Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, since the integrand is positive on $Q(0,2)$. Let $x \in Q(0,2)$ with $x \neq 0$, and let $r := |x|_\infty$, so that $0 < r \le 1$ and $r \le |x|$. The cube $Q(0,2r)$ contains x, is contained in $Q(0,2)$, where $f_\delta(y) = |y|^{\delta-d}$, and contains $B(0,r)$. The definition of M, together with bounding the marked factors, gives

$$\begin{aligned}
Mf_\delta(x) &\ge \frac{1}{(2r)^d}\underbrace{\int_{Q(0,2r)} f_\delta\,dy}_{\ge \int_{B(0,r)}|y|^{\delta-d}\,dy} \ge \frac{1}{(2r)^d}\underbrace{\int_{B(0,r)} |y|^{\delta-d}\,dy}_{=|S^{d-1}|\,r^\delta/\delta \text{ by (14)}} \\
&= \frac{|S^{d-1}|}{2^d\delta}\underbrace{r^{\delta-d}}_{\ge |x|^{\delta-d},\ \text{as } \delta-d<0} \ge \frac{|S^{d-1}|}{2^d\delta}\,f_\delta(x).
\end{aligned}$$

Since $f_\delta = 0$ outside $Q(0,2)$, this gives $Mf_\delta \ge \frac{|S^{d-1}|}{2^d\delta}\,f_\delta$ almost everywhere, and taking norms gives

$$\|Mf_\delta\|_{L^p_{w_\delta}} \ge D\,\delta^{-1}\,\|f_\delta\|_{L^p_{w_\delta}}, \qquad D := 2^{-d}\,|S^{d-1}|. \tag{16}$$

Applying (13) to w_δ and f_δ, and dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$, gives together with (16), (15) and $\alpha \ge 0$

$$D\,\delta^{-1} \le \frac{\|Mf_\delta\|_{L^p_{w_\delta}}}{\|f_\delta\|_{L^p_{w_\delta}}} \le c\,[w_\delta]_{A_p}^\alpha \le c\,C^\alpha\,\delta^{-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Hence

$$\delta^{\alpha(p-1)-1} \le \frac{c\,C^\alpha}{D} \qquad \text{for every } 0 < \delta < 1.$$

The assumption $\alpha < \frac{1}{p-1}$ makes the exponent $\alpha(p-1)-1$ negative, so the left-hand side tends to ∞ as $\delta \to 0$, while the right-hand side is a constant. This is a contradiction. □

A Proof of Proposition 1

Proof. For every cube Q and $x \in Q$, $Mf(x) \ge \frac{1}{|Q|}\int_Q |f|\,dy$, so (1) gives

$$\Big(\frac{1}{|Q|}\int_Q |f|\,dy\Big)^p w(Q) \le \int_Q (Mf)^p\,w\,dx \le C^p \int_{\mathbb{R}^d} |f|^p\,w\,dx \tag{17}$$

for every cube Q and every $f \in L^p_w(\mathbb{R}^d)$. Now let Q be a cube. Since w is not almost everywhere zero, Q is contained in a cube Q_0 with $w(Q_0) > 0$. Then (17) with Q_0 in place of Q and $f = \chi_Q$ gives $(|Q|/|Q_0|)^p\,w(Q_0) \le C^p\,w(Q)$, so $w(Q) > 0$.

For $n \ge 1$ let $f_n := \big(w + \frac{1}{n}\big)^{-\frac{1}{p-1}}\chi_Q$. Since $w \le w + \frac{1}{n}$,

$$|f_n|^p\,w \le \big(w+\tfrac{1}{n}\big)^{-\frac{p}{p-1}}\big(w+\tfrac{1}{n}\big) = f_n \le n^{\frac{1}{p-1}},$$

so $f_n \in L^p_w(\mathbb{R}^d)$, and $w < \infty$ almost everywhere gives $\int_Q f_n\,dy > 0$. Then (17) with $f = f_n$, together with $|f_n|^p w \le f_n$, gives

$$\Big(\frac{1}{|Q|}\int_Q f_n\,dy\Big)^p w(Q) \le C^p \int_Q f_n\,dy.$$

The functions f_n increase to $w^{-\frac{1}{p-1}}\chi_Q$ as $n \to \infty$, so dividing by $\int_Q f_n\,dy$ gives

$$\begin{aligned} C^p &\ge \lim_{n\to\infty} \frac{w(Q)}{|Q|}\Big(\frac{1}{|Q|}\int_Q f_n\,dy\Big)^{p-1} \\ &= \frac{w(Q)}{|Q|}\Big(\frac{1}{|Q|}\int_Q \lim_{n\to\infty} f_n\,dy\Big)^{p-1} \qquad \text{(monotone convergence)} \\ &= \frac{w(Q)}{|Q|}\Big(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\Big)^{p-1}. \end{aligned}$$

Since $w(Q) > 0$, $\int_Q w^{-\frac{1}{p-1}}\,dy < \infty$. As Q was arbitrary, $w > 0$ almost everywhere. □

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